

Toward Differentiable Query Planners: Evidence from Multi-Tenant Pipelines

Dilnoza Kowalczyk, Elke Vasquez

Calder Bay University

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Abstract

We examine the behaviour of streaming joins in heterogeneous clusters. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

1. Introduction

The reduced model retains most of the explanatory power at a fraction of the computational cost. The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters. Sensitivity to the smoothing window proved modest across the range we examined. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses.

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2. Related Work

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Longer observation windows narrow the confidence interval but do not shift the central estimate.

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Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification. The reduced model retains most of the explanatory power at a fraction of the computational cost. Replicate agreement was high, suggesting the protocol is robust to small handling differences. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol.

3. Materials and Methods

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4. Experimental Setup

The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses.

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5. Results

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sixty percent of nominal capacity.

6. Discussion

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Calibration drift over the trial period remained below the threshold set out in the protocol. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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7. Limitations

We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters.

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